UNIT 1

Problems based on discrete random variable

1) A discrete random variable has the following probability distribution.

x	0	1	2	3	4	5	6	7	8
$P(x)$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

i) Find the value of a.

ii) Find the prob. $P(x < 3)$

iii) Find $P(0 < x < 3)$

iv) Find $P(x \geq 3)$

v) Find the distributive function

vi) Find the mean value of x.

Soln :-

i) To find : a

W.K.T, $\sum P(x) = 1$

$$a + 3a + 5a + 7a + 9a + 11a + 13a + 15a + 17a = 1$$

$$81a = 1$$

$$a = \frac{1}{81}$$

ii) $P(x < 3)$

$$P(x < 3) = P(X = 0) + P(x = 1) + P(x = 2)$$

$$= a + 3a + 5a$$

$$= 9a = 9 \times \frac{1}{81} = \frac{1}{9}$$

iii) $P(0 < x < 3)$

$$P(0 < x < 3) = P(x=1) + P(x=2)$$
$$= 3a + 5a$$
$$= 8a = 8 \times \frac{1}{81}$$
$$= 8/81$$

iv) $P(x \geq 3)$

$$P(x \geq 3) = P(x=3) + P(x=4) + P(x=5) + P(x=6) + P(x=7) + P(x=8)$$
$$= 7a + 9a + 11a + 13a + 15a + 17a$$
$$= 72a = \frac{72}{81} = \frac{8}{9}$$

v) Distributive function (cummulative (+))

x	0	1	2	3	4	5	6	7	8
$P(x)$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

$F(x)$: cummulative addition

x,	0	1	2	3	4	5	6	7	8
$P(x)$	$\frac{1}{81}$	$\frac{3}{81}$	$\frac{5}{81}$	$\frac{7}{81}$	$\frac{9}{81}$	$\frac{11}{81}$	$\frac{13}{81}$	$\frac{15}{81}$	$\frac{17}{81}$
$F(x)$	$\frac{1}{81}$	$\frac{4}{81}$	$\frac{9}{81}$	$\frac{16}{81}$	$\frac{25}{81}$	$\frac{36}{81}$	$\frac{49}{81}$	$\frac{64}{81}$	1

v) To find : mean

$$E(x) = \sum x\, p(x)$$

$$= 0 \times p(0) + 1(P(1)) + 2(P(2)) + 3(P(3)) + 4(P(4)) + 5(P(5)) + 6(P(6) + 7(P(7) + 8(P(8))$$

$$= 0 \times \frac{1}{81} + 1 \times \frac{3}{81} + 2 \times \frac{5}{81} + 3 \times \frac{7}{81} + 4 \times \frac{9}{81} + 5 \times \frac{11}{81} + 6 \times \frac{13}{81} + 7 \times \frac{15}{81} + 8 \times \frac{17}{81}$$

$$= 5.48$$

vi) $P(x>2/x<6)$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{P(x>2 \cap x<6)}{P(x<6)}$$

$$= \frac{P(x=3)+P(x=4)+P(x=5)}{P(x=0)+P(x=1)+P(x=2)+P(x=3)+P(x=4)+P(x=5)}$$

$$= \frac{7a+9a+11a}{0+3a+5a+7a+9a+11a} = \frac{27a}{35a}$$

$$= \frac{2}{3}$$

a) A random variable x has the probability distribution :

x	0	1	2	3	4	5	6	7
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

i) Find k, a) $P(x < 6)$ b) $P(x \geq 6)$

c) $P(1 < x < 5)$ d) $P(x > 2)$

ii) Find $P(1.5 < x < 4.5 / x > 2)$

iii) Find the distribution of x and find the minimum value of λ if $P(x \leq \lambda) > \frac{1}{2}$

Soln :-

i) Find k :-

W.K.T, $\sum P(x) = 1$

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$9k + 10k^2 = 1$$

$$10k^2 + 9k - 1 = 0$$

$$k = 0.1, -1$$

In prob dont consider negative & greater than 1.

$$\therefore k = 0.1 = \frac{1}{10}$$

a) $P(x < 6)$

$= P(x=0) + P(x=1) + P(x=2) + P(x=3) +$

$P(x=4) + P(x=5)$

$= 0 + k + 2k + 2k + 3k + k^2$

$= 8k + k^2$

$= \frac{8}{10} + \frac{1}{100} = \frac{80+1}{100} = \frac{81}{100}$

b) $P(x \geq 6)$

$= P(x=6) + P(x=7)$

$= 2k^2 + 7k^2 + k$

$= 9k^2 + k$

$= 9 \times \frac{1}{100} + \frac{1}{10}$

$= \frac{9+10}{100} = \frac{19}{100}$

c) $P(1 < x < 3)$

$= P(x=2) + P(x=3) + P(x=4)$

$= 2k + 2k + 3k$

$= 7k = 7 \times 1/10 = 7/10$

d) $P(x > 2)$

$= P(x=3) + P(x=4) + P(x=5) + P(x=6) + P(x=7)$

$= 2k + 3k + k^2 + 2k^2 + 7k^2 + k$

$= 6k + 10k^2$

$= \frac{6}{10} + \frac{10}{100} = 7/10$

iii) $P(x<6/x\geq 4)$

$$P(x<6/x\geq 4) = \frac{P(4\leq x<6)}{P(x\geq 4)}$$

$$= \frac{P(x=4)+P(x=5)}{P(x=4)+P(x=5)+P(x=6)+P(x=7)}$$

$$= \frac{3a+a^2}{3a+a^2+2a^2+(7a^2+a)}$$

$$= \frac{3a+a^2}{10a^2+4a} = \frac{3(0.1)+0.01}{10\times 0.01+4(0.1)}$$

$$= \frac{0.31}{0.5} = \frac{31}{50} = 0.62$$

iv) Distributive function :-

x	0	1	2	3	4	5	6	7
$P(x)$	0	$1/10$	$2/10$	$2/10$	$3/10$	$1/100$	$2/100$	$17/1$
$F(x)$	0	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{5}{10}$	$\frac{8}{10}$	$\frac{81}{100}$	$\frac{83}{100}$	1

Min value of λ :

$$P(x \leq \lambda) > \frac{1}{2}$$

$$\therefore \lambda = 4$$

Bay's Theorem:-

Let $B_1, B_2, \ldots B_n$ be an exhaustive and mutually exclusive random experiments and A be an event related to that Bi then.

$$P(B_i/A) = \frac{P(B_i)\, P(A/B_i)}{\sum_{i=1}^{n} P(B_i)\, P(A/B_i)}, \quad i = 1, 2, 3, \ldots n$$

3) The contents of urn I, II, III are as follows: 1 white, 2 black and 3 red balls; 2 white, 1 black and 1 red balls; 4 white, 5 blacks and 3 red balls. One urn is chosen at random and 2 balls drawn from it. They happen to be white and red. What is the possibility that they come

Page :
Date : / /

from ums I, II and III ?

Soln :

Let E → getting 1 white and 1 Red

$$P(U_1/E) = \frac{P(U_1) \times P(E/U_1)}{P(U_1) \times P(E/U_1) + P(U_2)\,P(E/U_2) + P(U_3)\,P(E/U_3)}$$

$$P(U_1) = \frac{1}{3}, \quad P(U_2) = \frac{1}{3} \quad P(U_3) = \frac{1}{3}$$

$$= \frac{\frac{1}{3} \times \frac{{}^1C_1 \times {}^3C_1}{{}^6C_2}}{\frac{1}{3} \times \left(\frac{{}^1C_1 \times {}^3C_1}{{}^6C_2}\right) + \frac{1}{3} \times \frac{{}^2C_1 \times {}^1C_1}{{}^4C_2} + \frac{1}{3} \times \frac{{}^4C_1 \times {}^3C_1}{{}^{12}C_2}}$$

$$= \frac{{}^1C_1 \times {}^3C_1 / {}^6C_2}{\frac{1\times 3}{\frac{6\times 5}{2\times 1}} + \frac{2\times 1}{\frac{4\times 3}{2\times 1}} + \frac{4\times 3}{\frac{12\times 11}{2\times 1}}}$$

$$= \frac{1\times 3 / \frac{6\times 5}{2\times 1}}{\frac{1}{5} + \frac{1}{3} + \frac{2}{11}}$$

$$= \frac{1/5}{\frac{3 + 5\times 11 + 15}{5\times 3\times 11}} = \frac{1/5}{\frac{33+55+15}{5\times 3\times 11}} = \frac{33}{118}.$$

$$P(U_2/E) = \frac{P(U_2) \times P(E/U_2)}{P(U_1) \times P(E/U_1) + P(U_2)\,P(E/U_2) + P(U_3)\,P(E/U_3)}$$

$$= \frac{\frac{1}{3} \times \frac{{}^2C_1 \times {}^1C_1}{{}^4C_2}}{\frac{1}{3}\left(\frac{1\times 3}{\frac{6\times 5}{2\times 1}} + \frac{2\times 1}{\frac{4\times 3}{2\times 1}} + \frac{4\times 3}{\frac{12\times 11}{2}}\right)}$$

$$= \frac{\frac{2\times 1}{\frac{4\times 3}{2}}}{\frac{1}{5} + \frac{1}{3} + \frac{2}{11}} = \frac{1/3}{\frac{1}{5} + \frac{1}{3}\times\frac{2}{11}}$$

Page :
Date : / /

$$= \frac{Y_3}{\frac{118}{55 \times 3}} = \frac{55}{118}$$

$$P(U_3/E) = 1 - [P(U_1/E) + P(U_2/E)]$$

$$= 1 - \left[\frac{33}{118} + \frac{55}{118}\right]$$

$$= 1 - \frac{88}{118} = \frac{118 - 88}{118} = \frac{30}{118}.$$

4 Find MGF, mean and variance of binomial distribution.

i) MGF :-

W.K.T, $P(X = x) = {}^nC_x p^x q^{n-x}$; $x = 0, 1, \ldots, n$

$$M_x(t) = E[e^{tx}] = \sum_x e^{tx} p(x)$$

$$= \sum_{x=0}^{n} e^{tx} {}^nC_x p^x q^{n-x}$$

$$= \sum_{x=0}^{n} (e^t)^x {}^nC_x p^x q^{n-x}$$

$$= \sum_{x=0}^{n} (e^t p)^x {}^nC_x q^{n-x}.$$

W.K.T, $(a+b)^n = \sum {}^nC_x a^x b^{n-x}$

$$\therefore M_x(t) = \sum_{x=0}^{n} {}^nC_x (e^t p)^x q^{n-x}$$

$$= (e^t p + q)^n = (q + e^t p)^n$$

ii) To find : Mean

W.K.T, $$E(x) = \left[\frac{d}{dt} M_x(t)\right]_{t=0}$$

$$E(x) = \left[\frac{d}{dt} (q + e^t p)^n\right]_{t=0}$$

$$= \left[n (q + e^t p)^{n-1} [0 + e^t p)\right]_{t=0}$$

$$= \left[n e^t p (q + e^t p)^{n-1}\right]_{t=0}$$

$= np(q+p)^{n-1}$ $[p+q=1]$

$= np(1)^{n-1}$

$= np$

iii) To find variance :-

$$E(x^2) = \left[\frac{d^2}{dt^2} M_x(t)\right]_{t=0}$$

$$= \left[\frac{d}{dt}\left[npe^t(q+pe^t)^{n-1}\right]\right]_{t=0}$$

$$\boxed{\text{Var}(x) = E(x^2) - [E(x)]^2}$$

$v = uv + v'u$

$u = e^t$	$v = (q+pe^t)^{n-1}$
$u' = e^t$	$v' = (n-1)(q+pe^t)^{n-2} pe^t$

$$= \left[np\left[e^t(q+pe^t)^{n-1}\right] + \left[(n-1)pe^{2t}(q+pe^t)^{n-2}\right]\right]_{t=0}$$

$$= np\left[e^0(q+pe^0)^{n-1}\right] + \left[(n-1)pe^0(q+pe^0)^{n-2}\right]$$

$$= np\left[(p+q)^{n-1} + (n-1)p(p+q)^{n-2}\right]$$

$$= np\left[1 + (n-1)p(1)^{n-2}\right]$$

$$= np\left[1 + (n-1)p\right]$$

$$= np\left[1 + np - p\right]$$

$$= np + n^2p^2 - np^2$$

$$\text{Var}(x) = \cancel{n^2p^2} - np^2 + np - \cancel{(np)^2}$$

$$= np - np^2 = p(n - np)$$

$$= np(1 - p)$$

5. State and prove Memory less property of G.D.

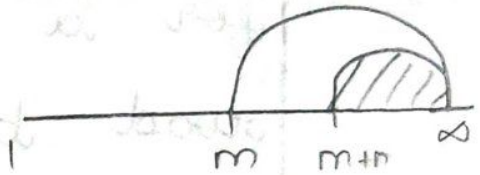
If X has a geometric distribution for any two positive integers m and n,

$$P[x > m+n / x > m] = P[x > n]$$

Proof :-

$$P[x > m+n / x > m] = P[x > n]$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$



$$\text{LHS} : \frac{P(x > m+n \cap x > m)}{P(x > m)}$$

$$= \frac{P[x > m+n]}{P(x > m)} \rightarrow (1)$$

$$P(X = x) = pq^{x-1}$$

$$P(x > k) = \sum_{k+1}^{\infty} pq^{x-1}$$

$$= pq^{k+1-1} + pq^{k+2-1} + pq^{k+3-1} + \dots$$

$$= pq^k + pq^{k+1} + pq^{k+2} + \dots$$

$$= pq^k + pq^k q^1 + pq^k q^2 + \dots$$

$$= pq^k (1 + q + q^2 + \dots)$$

$$= pq^k (1-q)^{-1} \quad \text{(binomial thm)}$$

$$= pq^k (p)^{-1}$$

$$= \frac{pq^k}{p}$$

$$= q^k$$

① $\Rightarrow$ LHS $= \dfrac{q^{m+n}}{q^m} = q^n \rightarrow$ (I)

RHS $= P(X > n) = q^n \rightarrow$ (II)

From (I) & (II),

LHS = RHS

$\therefore P(x > m+n / x > m) = P(x > n)$

6) If the density function of a continuous R.V x is given,

$$f(x) = \begin{cases} ax & ; 0 \le x \le 1 \\ a & ; 1 \le x \le 2 \\ 3a - ax & ; 2 \le x \le 3 \\ 0 & ; \text{otherwise} \end{cases}$$

i) Find a ii) Find CDF of x

i) To find : a

W.K.T, $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$$\int_0^1 ax\,dx + \int_1^2 a\,dx + \int_2^3 3a - ax\,dx = 1$$

$$a\int_0^1 x\,dx + a\int_1^2 dx + a\int_2^3 (3-x)\,dx = 1$$

$$a\left[\frac{x^2}{2}\right]_0^1 + a\left[x\right]_1^2 + a\left[3x - \frac{x^2}{2}\right]_2^3 = 1$$

$$\frac{a}{2}[1-0] + a[2-1] + a\left[\left(9 - \frac{9}{2}\right) - \left(6 - \frac{4}{2}\right)\right] = 1$$

$$\frac{a}{2} + a + a\left[\frac{9}{2} - 4\right] = 1$$

$$\frac{a}{2} + \frac{a^{x2}}{1\times 2} + \frac{a}{2} = 1$$

$$\frac{2a + 2a}{2} = 1 \quad = \frac{4a}{2} = 1$$

$$a = 1/2$$

$$\therefore f(x) = \begin{cases} x/2 & ; 0 \le x \le 1 \\ 1/2 & ; 1 \le x \le 2 \\ \frac{3}{2} - \frac{1}{2}x & ; 2 \le x \le 3 \\ 0 & ; \text{otherwise} \end{cases}$$

	0	$x/2$	$1/2$	$\frac{3}{2}-\frac{1}{2}x$	0	
$-\infty$	0	1	2	3	∞	

ii) To find : CDF

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(x)\,dx.$$

① $$F(x) = \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{x} f(x)\,dx.$$

$$= 0 + \int_{0}^{x} \frac{x}{2}\,dx = \frac{1}{2}\left[\frac{x^2}{2}\right]_0^x$$

$$= \frac{x^2}{4}$$

② For $1 \le x \le 2$,

$$F(x) = \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{1} f(x)\,dx + \int_{1}^{x} f(x)\,dx$$

$$= 0 + \frac{1}{2}\left[\frac{x^2}{2}\right]_0^1 + \frac{1}{2}[x]_1^x$$

$$= \frac{1}{4}(1-0) + \frac{1}{2}(x-1) = \frac{1}{4} + \frac{1}{2}x - \frac{1}{2}$$

$$= \frac{1}{4} - \frac{2}{4} + \frac{1}{2}x = \frac{1}{2\times 2}x - \frac{1}{4}$$

$$= \frac{2x-1}{4}$$

④ For $2 \le x \le 3$

$$F(x) = \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{1} f(x)\,dx + \int_{1}^{2} f(x)\,dx + \int_{2}^{x} f(x)\,dx$$

$$= \frac{1}{4} + \int_{1}^{2} \frac{1}{2}\,dx + \int_{2}^{x} \frac{3}{2} - \frac{1}{2}x\,dx$$

$$= \frac{1}{4} + \frac{1}{2}[x]_{1}^{2} + \frac{1}{2}\int_{2}^{x} (3-x)\,dx$$

$$= \frac{1}{4} + \frac{1}{2}[2-1] + \frac{1}{2}\left[3x - \frac{x^2}{2}\right]_{2}^{x}$$

$$= \frac{1}{4} + \frac{1}{2} + \frac{1}{2}\left[\left(3x - \frac{x^2}{2}\right) - \left(6 - \frac{4}{2}\right)\right]$$

$$= \frac{1+2}{4} + \frac{1}{2}\left[\left(3x - \frac{x^2}{2}\right) - 4\right]$$

$$= \frac{3}{4} + \frac{3x}{2} - \frac{x^2}{4} - \frac{2}{1}$$

$$= \frac{6x - x^2 - 8 + 3}{4} = \frac{6x - x^2 - 5}{4}$$

⑤ $x > 3$

$$F(x) = \frac{1}{4} + \frac{1}{2} + \frac{1}{2}\left[3x - \frac{x^2}{2}\right]_{2}^{3} + \int_{3}^{x} f(x)\,dx$$

$$= \frac{1+2}{4} + \frac{1}{2}\left[\left(9 - \frac{9}{2}\right) - \left(6 - \frac{4}{2}\right)\right] + 0$$

$$= \frac{3}{4} + \frac{1}{2}\left(9 - \frac{9}{2} - 6 + \frac{4}{2}\right)$$

$$= \frac{3}{4} + \frac{1}{4} = 1$$

$$\therefore \text{CDF}, \quad F(x) = \begin{cases} 0 & ; -\infty < x < 0 \\ \frac{x^2}{4} & ; 0 \le x \le 1 \\ \frac{2x-1}{4} & ; 1 \le x \le 2 \\ \frac{6x-x^2-5}{4} & ; 2 \le x \le 3 \\ 1 & ; x > 3 \end{cases}$$

UNIT 2

Problems :-

1) The joint prob. mass function of X, Y is gn by $P(x, y) = k(2x + 3y)$, $x = 0, 1, 2$ $y = 1, 2, 3$ Find all the marginal and conditional prob. distribution also find the prob. distribution of $(x+y)$ and $P(x+y>3)$. Examin x and Y are independent.

$P(x, y) = k(2x + 3y)$, $x = 0, 1, 2$ $y = 1, 2, 3$

x/y	0	1	2	Total
1	(0,1) $3k = 3/72$	(1,1) $5k = 5/72$	(2,1) $7k = 7/72$	$15k = 15/72$
2	(0,2) $6k = 6/72$	(1,2) $8k = 8/72$	(2,2) $10k = 10/72$	$24k = \frac{24}{72}$
3	(0,3) $9k = 9/72$	(1,3) $11k = 11/72$	(2,3) $13k = 13/72$	$33k = \frac{33}{72}$
Total	$18k = 18/72$	$24k = 24/72$	$30k = \frac{30}{72}$	$72k = 1$

i) $72k = 1$

$k = 1/72$

$\sum_j \sum_i P_{ij} = 1$

Sub in table,

ii) To find : marginal distribution x

x	0	1	2
P(x)	18/72	24/72	30/72

Marginal distribution of y :

y	1	2	3
P(y)	15/72	24/72	33/72

iii) Conditional distribution of X :-

$$P(x=0/y=1) = \frac{P(x=0, y=1)}{P(y=1)} = \frac{3/72}{15/72} = \frac{3}{15} = \frac{1}{5}$$

$$P(x=1/y=1) = \frac{P(x=1, y=1)}{P(y=1)} = \frac{5/72}{15/72} = \frac{5}{15} = \frac{1}{3}$$

$$P(x=2/y=1) = \frac{P(x=2, y=1)}{P(y=1)} = \frac{7/72}{15/72} = \frac{7}{15}$$

$$P(x=0/y=2) = \frac{P(x=0, y=2)}{P(y=2)} = \frac{6/72}{24/72} = \frac{1}{4}$$

$$P(x=1/y=2) = \frac{P(x=1, y=2)}{P(y=2)} = \frac{8/72}{24/72} = \frac{1}{3}$$

$$P(x=2/y=2) = \frac{P(x=2, y=2)}{P(y=2)} = \frac{10/72}{24/72} = \frac{5}{12}$$

$$P(x=0/y=3) = \frac{P(x=0, y=3)}{P(y=3)} = \frac{9/72}{33/72} = \frac{3}{11}$$

$P(Y=3/x=0) = \frac{P(x=0, y=3)}{P(x=0)} = \frac{9/72}{18/72} = \frac{1}{2}$

$$P(x=1/y=3) = \frac{P(x=1, y=3)}{P(y=3)} = \frac{11/72}{33/72} = \frac{1}{3}$$

$$P(x=2/y=3) = \frac{P(x=2, y=3)}{P(y=3)} = \frac{13/72}{33/72} = \frac{13}{33}$$

Conditional distribution of Y is,

$$P(Y=1/x=0) = \frac{P(x=0, y=1)}{P(x=0)} = \frac{3/72}{18/72} = \frac{1}{6}$$

$$P(Y=2/x=0) = \frac{P(x=0, y=2)}{P(x=0)} = \frac{6/72}{18/72} = \frac{1}{3}$$

$$P(Y=1/x=1) = \frac{P(x=1, y=1)}{P(x=1)} = \frac{5/72}{24/72} = \frac{1}{2}$$

$$P(Y=2/x=1) = \frac{P(x=1, y=2)}{P(x=1)} = \frac{8/72}{24/72} = \frac{1}{3}$$

$$P(Y=3/x=1) = \frac{P(x=1, y=3)}{P(x=1)} = \frac{11/72}{24/72} = \frac{11}{24}$$

$$P(Y=1/x=2) = \frac{P(x=2, y=1)}{P(x=1)} = \frac{7/72}{30/72} = \frac{7}{30}$$

$$P(Y=2/x=2) = \frac{P(x=2, y=2)}{P(x=2)} = \frac{10/72}{30/72} = \frac{1}{3}$$

$$P(Y=3/x=2) = \frac{P(x=2, y=3)}{P(x=2)} = \frac{13/72}{30/72} = \frac{13}{30}$$

iv) $(x+y)$

$x+y$	P
1) $P(0,1)$	$3/72$
2) $P(0,2) + P(1,1)$	$\frac{6}{72} + \frac{5}{72} = \frac{11}{72}$
3) $P(0,3) + P(2,1) + P(1,2)$	$\frac{9}{72} + \frac{8}{72} + \frac{7}{72} = \frac{24}{72}$
4) $P(1,3) + P(2,2)$	$\frac{11}{72} + \frac{10}{72} = \frac{21}{72}$
5) $P(2,3)$	$13/72$

v) $P(x+y > 3)$

$$= P(x+y=4) + P(x+y=5)$$

$$= \frac{21}{72} + \frac{13}{72} = \frac{34}{72}$$

vi) Verify independent :-

If $x+y$ is independent then

$$P[x=i] \times P[y=j] = P[i,j]$$

Now, $P(x=0) \times P(y=1) = \frac{18}{72} \times \frac{15}{72} = 0.05$

$$P(0,1) = \frac{3}{72} = 0.06$$

Hence, $P(x=0) \times P(y=1) \neq P(0,1)$

x and y are not independent.

2) Total distribution of x and y is gn by $f(x,y) = \frac{x+y}{21}$, $x = 1,2,3$, $y = 1,2$. Find margin distribution $E(XY)$

$f(x) = \frac{x+y}{21}$; $x = 1,2,3$ $y = 1,2$

x/y	1	2	Total
1	(1,1) 2/21	(1,2) 3/21	5/21
2	(2,1) 3/21	(2,2) 4/21	7/21
3	(3,1) 4/21	(3,2) 5/21	9/21
Total	9/21	12/21	= 1

Marginal distribution of x

X	1	2	3
P(x)	5/21	7/21	9/21

Marginal distribution of y

Y	1	2
P(Y)	9/21	12/21

$E(XY) = \sum\sum xy\, f(x,y)$

$x = 1, y = 1 \Rightarrow 1 \times 1 \times \frac{2}{21} = \frac{2}{21}$

$x = 1, y = 2 \Rightarrow 2 \times 1 \times \frac{3}{21} = 6/21$

$X = 2, \quad Y = 1 \Rightarrow 2 \times 3/21 = 6/21$

$X = 2, \quad Y = 2 \Rightarrow 4 \times 4/21 = 16/21$

$X = 3, \quad Y = 1 \Rightarrow 3 \times 4/21 = 12/21$

$X = 3, \quad Y = 2 \Rightarrow 6 \times 5/21 = 30/21$

$$E(XY) = \frac{2 + 6 + 6 + 16 + 21 + 30}{21}$$

$$= \frac{72}{21} = \frac{24}{7}$$

3) The joint pdf of random variable $f(x,y) = kxye^{-(x^2+y^2)}$, $x>0, y>0$, find the value of k, p.T X & Y are independent.

W.K.T, $\iint f(x,y)\,dx\,dy = 1$

$$\int_0^\infty\int_0^\infty kxye^{-(x^2+y^2)}\,dx\,dy = 1$$

$$k\int_0^\infty\int_0^\infty xye^{-(x^2+y^2)}\,dx\,dy = 1$$

$$k\int_0^\infty\int_0^\infty xe^{-x^2}ye^{-y^2}\,dx\,dy = 1$$

$$k\int_0^\infty xe^{-x^2}dx\int_0^\infty ye^{-y^2}dy = 1 \longrightarrow (1)$$

Evaluate:

$$= \int_0^\infty xe^{-x^2}dx$$

$$= \int_0^\infty e^{-t}\frac{dt}{2}$$

$$= \frac{1}{2}\int_0^\infty e^{-t}dt$$

$$= \frac{1}{2}\left[\frac{e^{-t}}{-1}\right]_0^\infty$$

$$= -\frac{1}{2}\left[e^{-\infty} - e^0\right]$$

$$= \frac{-1}{2}[0-1] = \frac{1}{2}$$

Put $x^2 = t$

$dt = 2x\,dx$

$\frac{dt}{2} = x\,dx$

Limit,

$x = 0 \Rightarrow x^2 = t$

$0 = t$

$x = \infty \Rightarrow \infty^2 = t$

$t = \infty$

Similarly,

$$\int_0^\infty ye^{-y^2}dy = \frac{1}{2}$$

Sub in ①,

$① \Rightarrow k(1/2)(1/2) = 1$

$k(1/4) = 1$

$k = 4$

ii) If x and Y are independent.

$f(x,y) = f(x) \cdot f(y)$

$f(x) = \int f(x,y)\,dy$

$= \int_0^\infty 4xye^{-(x^2+y^2)}\,dy$

$= \int_0^\infty 4xye^{-x^2-y^2}\,dy$

$= \int_0^\infty 4xye^{-x^2}e^{-y^2}\,dy$

$= 4xe^{-x^2}\int_0^\infty ye^{-y^2}\,dy$

$= 4xe^{-x^2}\left[\frac{1}{2}\right]$

$= 2xe^{-x^2}$

Illy, $f(y) = 2ye^{-y^2}$

Now, $f(x) \cdot f(y) = 2xe^{-x^2} \times 2ye^{-y^2}$

$= 4xye^{-x^2-y^2}$

$= 4xye^{-(x^2+y^2)}$

$= f(x,y)$

16m (X)

4) If X and Y are independent R.V with pdf e^{-x} ; $x \geq 0$; e^{-y} ; $y \geq 0$. Find the probability function of $U = \frac{X}{X+Y}$, $V = X+Y$. are U, V independent?

Gn : $f(x) = e^{-x}$; $x \geq 0$

$f(y) = e^{-y}$; $y \geq 0$

Since X & Y are independent,

$$f(x,y) = f(x)f(y)$$
$$= e^{-x}e^{-y} = e^{-x-y}$$
$$= e^{-(x+y)} \quad ; \quad x \geq 0, y \geq 0$$

Gn : $U = \frac{X}{X+Y}$; $V = X+Y$

$u = \frac{x}{x+y}$; $v = x+y$.

$u = \frac{x}{v}$ | $v = uv + y$

$x = uv$ | $y = v - uv$

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$\frac{\partial x}{\partial u} = v \; ; \; \frac{\partial x}{\partial v} = u.$$

$$\frac{\partial y}{\partial u} = -v \; ; \; \frac{\partial y}{\partial v} = 1-u.$$

$$J = \begin{vmatrix} v & u \\ -v & (1-u) \end{vmatrix}$$

$$= v(1-u) + uv$$

$$= v - \cancel{uv} + \cancel{uv}$$

$$J = v$$

$$f_{xy}(u,v) = e^{-(x+y)}$$

$$= e^{-(uv+v-uv)}$$

$$= e^{-v} \longrightarrow ①$$

$$f_{uv}(u,v) = |J| f_{xy}(u,v)$$

$$= ve^{-v} \longrightarrow ②$$

Gn: $x \geq 0$ | $y \geq 0$

$uv \geq 0$ | $v - uv \geq 0 \Rightarrow v(1-u) \geq 0$

$u \geq 0 ; v \geq 0$ | $v \geq 0 ; (1-u) \geq 0$

$0 \leq u ; 0 \leq v$ | $v \geq 0 ; 1 \geq u$

$u \leq 1$

$$\boxed{0 \leq u \leq 1 \; ; \; 0 \leq v}$$

$f(u) = \int_0^\infty ve^{-v} dv \Rightarrow u = v \quad | \quad dv = e^{-v} dv$

$du = dv \quad | \quad v = -e^{-v}$

$$f(u) = [-ve^{-v}]_0^\infty - \int_0^\infty (-e^{-v}) dv$$

$$= (0-0) + (-e^{-v})_0^\infty$$

$$= 0 - (0-1)$$

$$= 1$$

$f(u) = 1 \; ; \quad 0 \le u \le 1$

To find : pdf.

$$f(v) = \int_{-\infty}^{\infty} f(u,v)\, du$$

$$f(v) = \int_0^1 ve^{-v} du$$

$$= ve^{-v} [u]_0^1$$

$$= ve^{-v} [1-0]$$

$\therefore f(v) = ve^{-v} \; ; \quad v \ge 0$

To check independence :-

$$f(u,v) = f(u) \cdot f(v)$$

RHS $= (1) \times (ve^{-v}) = ve^{-v}$

LHS $= f(u,v) = ve^{-v}$

$\therefore$ LHS = RHS

U and V are independent

H.W 5) If 2 R.V of X and Y having JPDF $f(x,y) = \begin{cases} 2-x-y, & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{elsewere.} \end{cases}$ find correlation Coefficient

Soln:

Marginal p.D.F of X

$$f(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$$

$$= \int_0^1 (2-x-y)\,dy$$

$$= \left(\frac{3}{2} - x\right),\ 0 \le x \le 1$$

Marginal PDF of Y

$$f(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

$$= \int_0^1 (2-x-y)\,dx$$

$$= \left(\frac{3}{2} - y\right),\ 0 \le x \le 1$$

$$E(X) = \int_0^1 x\left(\tfrac{3}{2} - x\right) dx \qquad E(Y) = \int_0^1 y\left(\tfrac{3}{2} - y\right) dy$$

$$= 5/12 \qquad = \frac{5}{12}$$

$$E[XY] = \int_0^1 \int_0^1 xy\,(2-x-y)\,dx\,dy$$

$$= \frac{1}{6}$$

$$\text{Cov}(X,Y) = E(XY) - E(X)E(Y)$$

$$= \frac{1}{6} - \frac{5}{12} \times \frac{5}{12} = -0.0069$$

$$E(X^2) = \int_0^1 x^2\left(\frac{3}{2} - x\right) dx.$$

$$= \int_0^1 \left(\frac{3x^2}{2} - x^3\right) dx$$

$$= \left[\frac{3x^3}{6_2} - \frac{x^4}{4}\right]_0^1$$

$$= \left[\frac{1}{2} - \frac{1}{4}\right] = \frac{1}{4}$$

Page :
Date : / /

$$E[Y^2] = \int_0^1 y^2\left(\frac{3}{2} - y\right) dy$$

$$= \int_0^1 \left(\frac{3y^2}{2} - y^3\right) dy$$

$$= \left[\frac{3}{2}\frac{y^3}{3} - \frac{y^4}{4}\right]_0^1$$

$$= \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$\sigma_x = \sqrt{E(X^2) - (E(X))^2} = \sqrt{\frac{1}{4} - \left(\frac{5}{12}\right)^2}$$

$$= \sqrt{\frac{1}{4} - \frac{25}{144}}$$

$$= \sqrt{0.25 - 0.1736}$$

$$= 0.27638$$

$$\sigma_y = \sqrt{E(Y^2) - (E[Y])^2} = \sqrt{\frac{1}{4} - \left(\frac{5}{12}\right)^2}$$

$$= 0.27638$$

$$\therefore \gamma(x,y) = \frac{\text{Cov}(X,Y)}{\sigma_x \sigma_y} = \frac{-0.0069}{0.27638 \times 0.2768}$$

$$= -0.0903$$

UNIT 4

1) Two horses A and B were tested according to their kind (in sec) to run a particular race with the following results :

A	28	30	32	33	33	29	34
B	29	30	30	24	27	29	-

Test whether horse A is running faster than B at 5% level.

Gn : $n_1 = 7$; $n_2 = 6$

$$\bar{x}_1 = \frac{28+30+32+33+33+29+34}{7}$$

$$= 31.28$$

$$\bar{x}_2 = \frac{29+30+30+24+27+29}{6}$$

$$= 28.16$$

$(x_1-\bar{x}_1)$	$(x_1-\bar{x}_1)^2$	$(x_2-\bar{x}_2)$	$(x_2-\bar{x}_2)^2$
-3.28	10.75	-0.83	0.68
-1.28	1.63	-0.17	0.02
0.72	0.51	-0.17	0.02
1.72	2.95	-4.16	17.30
1.72	2.95	2.83	8
-2.28	5.19	0.83	0.68
2.72	7.39	-	-
Σ =	31.37	Σ =	26.7

$$\therefore s^2 = \frac{\Sigma (x_1 - \bar{x}_1)^2 + \Sigma (x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}$$

$$= \frac{31.37 + 26.7}{7+6-2} = \frac{58.07}{11}$$

$$s^2 = 5.28$$

$$\therefore s = 2.29$$

Step 1:

$(\mu_1 = \mu_2)$

$H_0 \rightarrow$ There is no significant diff

$H_1 \rightarrow$ There is significant diff

Step 2: $(\mu_1 > \mu_2)$ [A horse is faster]

$$\alpha = 0.05$$

$$d.f = n_1 + n_2 - 2 = 11$$

Step 3:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$= \frac{31.28 - 28.16}{\sqrt{5.28\left(\frac{1}{7} + \frac{1}{6}\right)}} = \frac{3.12}{\sqrt{5.28\left(\frac{13}{42}\right)}}$$

$$= \frac{3.12}{\sqrt{5.28 \times 0.30}} = \frac{3.12}{\sqrt{1.584}} = 2.47$$

$$|t| = 2.47$$

Step 4 : table value

α with d.f $\Rightarrow$ (0.05, 11)

$t_\alpha = 1.796$

Step 5 : Conclusion

$t = 2.47 > t_\alpha = 1.796$

$\therefore$ Reject H_0

$\therefore$ Horse A runs faster than B.

(X) 21 A random sample of 10 boys had the following IQs : 70, 120, 110, 101, 88, 83, 95, 98, 107, 100

Do these data support the assumption of a population mean IQ of 100? Find the reasonable range in which most of the mean IQ values of samples of 10 boys lie ?

Gn : $n = 10$

$\mu = 100$

$$\bar{x} = \frac{70 + 120 + 110 + 101 + 88 + 83 + 95 + 98 + 107 + 100}{10}$$

$\therefore \bar{x} = 97.2$

$$s^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1}$$

x	$x - \bar{x}$	$(x-\bar{x})^2$
70	-27.2	739.84
120	22.8	519.84
110	12.8	163.84
101	3.8	14.44
88	-9.2	84.64
83	-14.2	201.64
95	-2.2	4.84
98	0.8	0.64
107	9.8	96.04
100	2.8	7.84

$$\therefore \quad \Sigma (x-\bar{x})^2 = 1833.6$$

$$S^2 = \frac{1833.6}{10-1} = 203.7$$

$$S = \sqrt{203.7} = 14.27$$

Step 1 :-

$H_0 \rightarrow$ There is no significant diff

$$\mu = 100$$

$H_1 \rightarrow$ There is significant diff

$$\mu \neq 100$$

Step 2 :

$\alpha = 0.05$

$d.f = n-1$

$= 10-1 = 9$

Step 3 :-

$$t = \frac{\bar{x}-\mu}{s/\sqrt{n}}$$

$$= \frac{97.2-100}{14.27/\sqrt{10}} = \frac{-2.8}{4.51}$$

$t = -0.62$

$\therefore |t| = 0.62$

Step 4 :-

t_α with d.f $(0.05, 9)$

$t_\alpha = 2.262$

Step 5 : conclusion

$t = 0.62 < t_\alpha = 2.262$

$\therefore$ Accept H_0

W.K.T, Range $= \bar{x} \pm t_\alpha \frac{s}{\sqrt{n}}$

Range $= 97.2 \pm 2.262 \times \frac{14.27}{\sqrt{10}}$

$= 97.2 \pm 10.20$

$= (97.2+10.20, 97.2-10.20)$

$= (107.4, 87) \Rightarrow (87, 107.4)$

3) A simple sample of heights 6400 english men has a mean of 170 cm and S.D = 6.4 cm, while a simple sam of heights of 1600 americans has a mean of 172 cm and S.D = 6.3 cm. Do the data indicate about americans or the average taller than english men.

Gn: $n_1 = 6400$, $\bar{x}_1 = 170$, $S_1 = 6.4$

$n_2 = 1600$, $\bar{x}_2 = 172$, $S_2 = 6.3$

Step 1:

$H_0: \mu_2 = \mu_1$

$H_1: \mu_1 < \mu_2$

Step : 2

$\alpha = 0.05$

Step : 3

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$= \frac{170 - 172}{\sqrt{\frac{6.4^2}{6400} + \frac{6.3^2}{1600}}} = \frac{-2}{0.104} = -19.23$$

$|Z| = 19.23$

Step 4:

$|Z|\alpha = 1.645$

Step 5:

$Z\alpha = 1.645 < Z = 19.23$

$\therefore$ Reject H_0

4) A group of 10 rats fed on diet A and another group of 8 rats fed on diet B recorded the following increase in weight

Diet A	5	6	8	1	12	4	3	9	6	10
Diet B	2	3	6	8	10	1	2	8		

Find if the variances are significantly different.

Gn: $n_1 = 10$; $n_2 = 8$

$$\bar{x}_1 = \frac{5+6+8+1+12+4+3+9+6+10}{10}$$

$$= 6.4$$

$$\bar{x}_2 = \frac{2+3+6+8+10+2+1+8}{8}$$

$$= 5$$

$(x_1-\bar{x}_1)$	$(x_1-\bar{x}_1)^2$	$(x_2-\bar{x}_2)$	$(x_2-\bar{x}_2)^2$
-1.4	1.96	-3	9
-0.4	0.16	-2	4
1.6	2.56	1	1
-5.4	29.16	3	9
5.6	31.36	5	25
-2.4	5.76	-4	16
-3.4	11.56	-3	9
2.6	6.76	3	9
-0.4	0.16	-	-
3.6	12.96	-	-
	102.4		82

$$S_1^2 = \frac{\Sigma (x_1-\bar{x}_1)^2}{n_1-1} = \frac{102.4}{9} = 11.37$$

$$S_2^2 = \frac{\Sigma (x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{82}{7} = 11.71$$

Step 1:

$H_0 \Rightarrow$ There is no significance diff

$$\sigma_1^2 = \sigma_2^2$$

$H_1 \Rightarrow$ There is significance diff

$$\sigma_1^2 \neq \sigma_2^2$$

Step 2:

$$\alpha = 0.05$$

$$df = (n_1 - 1, n_2 - 1) = (9, 7)$$

Step 3:

$$F = \frac{S_1^2}{S_2^2} \Rightarrow \frac{S_2^2}{S_1^2} = \frac{11.71}{11.37} = 1.02$$

Step 4:

$$\alpha = 0.05, \quad d.f = (7, 9)$$

$$F_\alpha = 3.68$$

Step 5: conclusion

$$F_\alpha = 3.68 > F = 1.02$$

$\therefore$ Accept H_0

5) 4 coins were tossed 160 times and the following results were obtained :

No. of heads	0	1	2	3	4
Observed frequencies	17	52	54	31	6

$n = 4$, $N = 160$

$P(X = x) = {}^nC_x \, p^n q^{n-x}$, $x = 0, 1, 2, 3, 4$

As $N = 160$ multiple with N,

$P[X = x) = N \times {}^nC_x \, p^n q^{n-x}$, $x = 0, 1, 2, 3, 4$

Probability of getting heads,

$p = 1/2$, $q = 1/2$

$P(X = x) = 160 \times {}^4C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x}$

$P(X = 0) = 160 \times {}^4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4$

$= 160 \times 1 \times 1 \times \frac{1}{16} = 10$

$P(X = 1) = 160 \times {}^4C_1 \times \left(\frac{1}{2}\right)^1 \times \left(\frac{1}{2}\right)^3$

$= \cancel{160}^{10} \times 4 \times \frac{1}{\cancel{2}} \times \frac{1}{\cancel{8}} = 40$

$P(X = 2) = 160 \times {}^4C_2 \times \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^2$

$= \cancel{160}^{20} \times \frac{\cancel{4} \times 3}{1 \times \cancel{2}} \times \frac{1}{\cancel{4}} \times \frac{1}{\cancel{4}}$

$= 60$

$$P(X=3) = 160 \times {}^4C_3 \times \left(\frac{1}{2}\right)^3 \times \left(\frac{1}{2}\right)^1$$

$$= \overset{40}{\cancel{160}} \times \frac{\cancel{4} \times 3 \times 2}{1 \times \cancel{2} \times \cancel{3}} \times \frac{1}{\cancel{8}_{\cancel{2}}} \times \frac{1}{\cancel{2}} = 40$$

$$P(X=4) = 160 \times {}^4C_4 \times \left(\frac{1}{2}\right)^4 \times \left(\frac{1}{2}\right)^0$$

$$= \overset{10}{\cancel{160}} \times 1 \times \frac{1}{\cancel{16}} \times 1 = 10$$

i) H_0 ⇒ There is no significance difference

∴ The event is unbiased

H_1 ⇒ There is significance difference

∴ The event is unbiased

ii) Level of significance :

$\alpha = 5\% = 0.05$

$d.f = n-1 = 5-1 = 4$

iii) Formula :-

O	E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
17	10	49	4.9
52	40	144	3.6
54	60	36	0.6
31	40	81	2.03
6	10	16	1.6
		Σ =	12.725

iv) Table values :-

χ^2_α with d.f = 4

$\chi^2_\alpha = 9.488$

v) Conclusion :-

$\chi^2 = 12.725 > \chi^2_\alpha = 9.488$

$\therefore$ Reject H_0

5. The following data are collected on two characters

	Smokers	Non-Smokers
Literates	83	57
Illiterates	45	68

Based on this, can you say that there is no relationship between smoking and literacy? Use χ^2 method.

UNIT 5

One way

1) The following table shows the lives in hours of 4 brands of electric lamps:

A	1610	1610	1650	1680	1700	17 0	1810	–
B	1580	1640	1640	1700	1750	–	–	–
C	1460	1550	1600	1620	1640	1660	1740	1820
D	1510	1520	1530	1570	1600	1680	–	–

Perform an analysis of variance, test the homogeneity of the mean lives the 4 brands of lamps.

Subtract 1600 and divide it by 10.

X_1	X_2	X_3	X_4
1	-2	-14	-9
1	4	-5	-8
5	4	0	-7
8	10	2	3
10	15	4	0
12.	–	6	8
20	–	14	–
–	–	22	–

X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
1	-2	-14	-9	-24	1	4	196	81
1	4	-5	-8	-8	1	16	25	64
5	4	0	-7	2	25	16	0	49
8	10	2	-3	17	64	100	4	9
10	15	4	0	29	100	225	16	0
12	-	6	8	26	144	-	36	64
20	-	14	-	34	400	-	196	-
-	-	22	-	22	-	-	484	-
Σ = 57	31	29	-19	98	735	361	957	267

① $H_0 \rightarrow$ There is no significant diff between the 4 brands of lamps

$H_1 \rightarrow$ There is significant difference between the 4 brands of lamps

Step 2 : Find N

No. of value gn in table (X_1, X_2, X_3, X_4)

$N = 26$

Step 3 : Total ans

$T = 98$

Step 4 : correct factor

$$C.F = \frac{T^2}{N} = \frac{(98)^2}{26} = 369.38$$

Step 5 : Total sum of square

$$TSS = \Sigma X_1^2 + \Sigma X_2^2 + \Sigma X_3^2 + \Sigma X_4^2 - \frac{T^2}{N}$$

$$= 735 + 361 + 957 + 267 - 369.38$$

$$= 1950.62$$

Step 6 : Find SSC

$$SSC = \frac{(\Sigma X_1)^2}{N_1} + \frac{(\Sigma X_2)^2}{N_2} + \frac{(\Sigma X_3)^2}{N_3} + \frac{(\Sigma X_4)^2}{N_4} - \frac{(\Sigma X)^2}{N}$$

$$= \frac{(57)^2}{7} + \frac{(31)^2}{5} + \frac{(29)^2}{8} + \frac{(-19)^2}{6} - 369.38$$

$$= 452.25$$

Step 7 : Sum of square btw error

$$SSE = TSS - SSC$$

$$= 1950.62 - 452.25$$

$$= 1498.37$$

Step 8 : ANOVA table c = no. of columns = 4

Source of variations	Sum of squares	DE	Mean sum of square	Variance ratio
b/w column	SSC = 452.25	C-1 = 4-1 = 3	MSC = $\frac{SSC}{C-1}$ = $\frac{452.25}{3}$ = 150.75	F = $\frac{MSE}{MSE}$ = $\frac{68.10}{150.75}$ = $\frac{150.75}{68.10}$ = 2.213
within column	SSE = 1498.37	N-C = 26-4 = 22	MSE = $\frac{SSE}{N-C}$ = $\frac{1498.37}{22}$ = 68.10	

Step 9 : Table value

Num ⇒ first , denom ⇒ 2nd

$F_c(3, 22) = 3.05$

Step 10 : Conclusion

$F = 2.213 < F_c(3, 22) = 3.05$

We accept H_0

2) There are three main brands of a certain powder. A set of 120 sample values is examined and found to be allocated among four groups (A, B, C and D) and three brands (I, II, III) as shown here under:

Brands	Groups			
	A	B	C	D
I	0	4	8	15
II	5	8	13	6
III	8	19	11	13

Is there any significant difference in brands preference? Answer at 5% level.

Soln

H_0 : There is no significant difference between brands

H_1 : There is a significant difference between brands.

Brands	Groups				Total	X_1^2	X_2^2	X_3^2	X_4^2
	A (X_1)	B (X_2)	C (X_3)	D (X_4)					
I (Y_1)	0	4	8	15	$\Sigma Y_1 = 27$	0	16	64	225
II (Y_2)	5	8	13	6	$\Sigma Y_2 = 32$	25	64	169	36
III (Y_3)	8	19	11	13	$\Sigma Y_3 = 51$	64	361	121	169
	$\Sigma X_1 = 13$	$\Sigma X_2 = 31$	$\Sigma X_3 = 32$	$\Sigma X_4 = 34$	$T = 110$	$\Sigma X_1^2 = 89$	$\Sigma X_2^2 = 441$	$\Sigma X_3^2 = 354$	$\Sigma X_4^2 = 430$

Step(1) $N = 12$

Step(2) $T = 110$

Step(3) $T^2/N = \frac{(110)^2}{12} = 1008.33$

Step(4)

$$TSS = \sum x_1^2 + \sum x_2^2 + \sum x_3^2 + \sum x_4^2 - \frac{T^2}{N}$$

$$= 89 + 441 + 354 + 430 - 1008.33$$

$$= 305.67.$$

Step(5)

$$SSR = \frac{(\sum y_1)^2}{N_2} + \frac{(\sum y_2)^2}{N_2} + \frac{(\sum y_3)^2}{N_2} - \frac{T^2}{N}$$

(Here N_2 = no of elts in each row)

$$= \frac{(27)^2}{4} + \frac{(32)^2}{4} + \frac{(51)^2}{4} - 1008.33$$

$$= 182.25 + 256 + 650.25 - 1008.33$$

$$= 80.17$$

Step (6)

$$SSE = TSS - SSR = 305.67 - 80.17 = 225.5$$

Step(7) ANOVA TABLE.

Source of Variation	SS	d.f	MS	Variance ratio	Table value at 5%
Between rows	SSR = 80.17	$r-1 = 3-1 = 2$	$MSR = \frac{SSR}{r-1} = \frac{80.17}{2} = 40.085$	$F_R = \frac{MSR}{MSE} = \frac{40.08}{20.06} = 1.99$	$F_R(2,9) = 4.26$
Error	SSE = 225.5	$N-r = 12-3 = 9$	$MSE = \frac{SSE}{N-r} = \frac{225.5}{9} = 20.06$		
Total	305.67				

Conclusion:-

If cal F_R < table F_R, we accept H_0.

i.e. $1.99 < 4.26$ is True.

So we accept H_0.

3) In order to compare three burners A, B & C an observation is made on each burner on three successive days. The data tabulated below.

	A	B	C
Day-1	6	28	14
Day-2	28	36	38
Day-3	8	32	20

Perform an analysis of variance on these data and find whether the difference between (i) the days and (ii) the burner's significant at 5% level of significance.

Solve by 2 way

4. Five doctors each test five treatments for a certain disease and observe the number of days each patient takes to recover. The recovery times in days are as follows.

Doctors	1	2	3.	4	5.
A	10	14	23	19	20
B	11	15	24	17	21
C	9	11	20	16	19
D	8	13	17	17	20
E	12	15	19	15	22

Is there any significant difference between the doctors and their treatments?

Solve by 2way

5) Analyze the variance in the following Latin square yields (in kg) of paddy where A, B, C, D denotes the different cultivation method.

S122	P121	R123	Q122
Q124	R123	P122	S125
P120	Q119	S120	R121
R122	S123	Q121	P122

Examine whether the different methods of cultivation have given significantly in its yields.

Rewrite the table,

Subtract by 120,

-S2	-P1	-R3	-Q2
-Q4	-R3	-P2	-S5
P0	Q1	S0	-B1
-R2	-S3	-Q1	-P2

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
	-2	-1	-3	-2	-8	4	1	9	4
	-4	-3	-2	-5	-14	16	9	4	25
	0	1	0	-1	0	0	1	0	1
	-2	-3	-1	-2	-8	4	9	1	4
Σ	-8	-6	-6	-10	-30	24	20	14	34

Step 1:

$H_0 \rightarrow$ There is no significant diff

$H_1 \rightarrow$ There is significant diff

Step 2:

Find T

$T = -30$

Step 3: Find N

$N = 16$

Step 4: Find CF

$$CF = \frac{T^2}{N} = \frac{(-30)^2}{16} = 56.25$$

Step 5 : Find TSS

$$TSS = \Sigma x_1^2 + \Sigma x_2^2 + \Sigma x_3^2 + \Sigma x_4^2 - CF$$

$$= 24 + 20 + 14 + 34 - 56.25$$

$$= 92 - 56.25$$

$$= 35.75$$

6) Find SSC :-

$$SSC = \frac{(\Sigma x_1)^2}{N_1} + \frac{(\Sigma x_2)^2}{N_1} + \frac{(\Sigma x_3)^2}{N_1} + \frac{(\Sigma x_4)^2}{N_1} - CF$$

$$= \frac{(-8)^2 + (-6)^2 + (-6)^2 + (-10)^2}{4} - 56.25$$

$$= 8.75$$

7) Find SSR :-

$$SSR = \frac{(-8)^2 + (-14)^2 + 0 + (-8)^2}{4} - 56.25$$

196

$$= 24.75$$

8) Find SSE :-

	P	Q	R	S
	0	-4	-2	-2
	-1	+1	-3	-3
	-2	-1	-3	-0
	-2	-2	-1	-5
Σ =	-5	-6	-9	-10

$$SSK = \frac{(-5)^2+(-6)^2+(-9)^2+(-10)^2}{4} - 56.25$$

$$= 4.25$$

9) Find SSE :-

$$SSE = TSS - SSC - SSR - SSK$$

$$= 35.75 - 2.75 - 24.75 - 4.25$$

$$= 4$$

10) ANOVA Table :-

Source of variation	Sum of squares	DF	Mean sum of squares	Variance ratio
b/w columns	SSC = 2.75	k-1=3	MSC = $\frac{2.75}{3}$ = 0.91	$F_C = \frac{0.91}{0.6}$ = 1.51
b/w rows	SSR = 24.75	k-1=3	MSR = $\frac{24.75}{3}$ = 8.25	$F_R = \frac{8.25}{0.6}$ = 13.75
b/w treatments	SSK = 4.25	3	MSK = $\frac{4.25}{3}$ = 1.41	$F_K = \frac{1.41}{0.6}$ = 2.35
Error	SSE = 4	6	MSE = $\frac{4}{6}$ = 0.6	

10) Table value :-

$F_C(3,6) = 4.76$, $F_R(3,6) = 4.76$, $F_K(3,6) = 4.76$

12) Conclusion :-

$F_C(3,6) = 4.76 > F_C = 1.51$ (∴ Accept H_0)

$F_R(3,6) = 4.76 < F_R = 13.75$ (does not accept H_0)

$F_K(3,6) = 4.76 > F_K = 2.35$ (∴ Accept H_0)

6) Analyze the variance in the following Latin square yields in kg of paddy when A, B, C, D denote the different cultivation methods.

B20	C17	D25	A34
A23	D21	C15	B24
D24	A26	B21	C19
C26	B23	A27	D22

Examine whether the different methods of cultivation have gn significantly different in yields

Step 1 :

H_0 → There is no significant difference between column, row and treatment.

H_1 → There is significant different b/w row, column and treatment

Step 2 : Find N

$N = 16$

Step 3 : Draw table and find T

Subtract by 25

-5^B	-8^C	0^D	9^A
-2^A	-4^D	-10^C	-1^B
-1^D	1^A	-4^B	-6^C
1^C	-2^B	2^A	-3^D

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	-5	-8	0	9	-4	25	64	0	81
Y_2	-2	-4	-10	-1	-17	4	16	100	1
Y_3	-1	1	-4	-6	-10	1	1	16	36
Y_4	1	-2	2	-3	-2	1	4	4	9
$\xi =$	-7	-13	-12	-1	-33	31	85	120	127

$T = -33$

Step 4 : Find CF

$$CF = \frac{T^2}{N} = \frac{(-33)^2}{16} = 68.06$$

Step 5 : Find TSS

$$TSS = \Sigma X_1^2 + \Sigma X_2^2 + \dots - CF$$

$$= 31 + 85 + 120 + 127 - 68.06$$

$$= 294.94$$

Step 6 : Find SSC

$$SSC = \frac{(\Sigma X_1)^2}{N_1} + \frac{(\Sigma X_2)^2}{N_1} + \dots - CF$$

$$= \frac{(-7)^2}{4} + \frac{(-13)^2}{4} + \frac{(-12)^2}{4} + \frac{(-1)^2}{4} - 68.0$$

$$= \frac{49 + 169 + 144 + 1}{4} - 68.06$$

$$= 90.75 - 68.06$$

$$= 22.69$$

⑦ Find SSR :

$$SSR = \frac{(\Sigma Y_1)^2}{N_2} + \frac{(\Sigma Y_2)^2}{N_2} + \dots - CF$$

$$= \frac{(-4)^2}{4} + \frac{(-17)^2}{4} + \frac{(-10)^2}{4} + \frac{(-2)^2}{4} - CF$$

$$= \frac{16 + 289 + 100 + 4}{4} - 68.06$$

$$= 34.19$$

Step 8 : Find SSK

A	B	C	D
-2	-5	1	-1
1	-2	-8	-9
2	-4	-10	0
9	1	-6	-3
10	-12	-23	-8

$$SSK = \frac{(10)^2 + (-12)^2 + (-23)^2 + (-8)^2}{4} - 68.06$$

$$= \frac{100 + 144 + 529 + 64}{4} - 68.06$$

$$= \frac{837}{4} - 68.06 = 141.19$$

Step 9: Find SSE

$$SSE = TSS - SSC - SSR - SSK$$

$$= 294.94 - 22.69 - 34.19 - 141.19$$

$$= 96.87$$

Step 10: ANOVA Table :-

Source of variation	Sum of squares	DF	Mean sum of squares	Variance ratio
b/w column	SSC = 22.69	C-1 = 3	$\frac{SSC}{C-1} = \frac{22.69}{3} = 7.56$	$F_C = \frac{16.15}{7.56} = 2.13$
b/w rows	SSR = 34.19	3	$\frac{34.19}{3} = 11.40$	$F_R = \frac{16.15}{11.40} = 1.41$
b/w treatments	SSK = 141.19	3	$\frac{141.19}{3} = 47.06$	$F_K = \frac{47.06}{16.15} = 2.91$
Error	SSE = 96.87	(4-1)(4-2) = 6	$\frac{96.87}{6} = 16.15$	

Step 11 : Table values

$F_C(6,3) = 8.94$

$F_R(6,3) = 8.94$

$F_T(3,6) = 4.76$

(Check ANOVA table when MSE is on num then its DF is written else the MSR/MSC's DF is written 1st.)

Step 12 : conclusion

$F_C(6,3) = 8.94 > F_C = 2.13$

$\therefore$ Accept H_0

$F_R(6,3) = 8.94 > F_R = 1.41$

$\therefore$ Accept H_0

$F_T(3,6) = 4.76 > F_T = 2.91$

$\therefore$ Accept H_0